

Q.1 In a cloud-based application where scalability is crucial, which cloud storage option allows dynamic scaling without the need for manual intervention?

Options:

- a. Block storage
- b. File storage
- c. Object storage
- d. Table storage

Correct Answer:

- c. Object storage

One-line Explanation:

Object storage automatically scales as data grows, making it ideal for highly scalable cloud applications.

Q.2 A mobile application needs to retrieve a list of all available restaurants from a food delivery service's API. The application should only read data without making any changes to the server resources. Which HTTP method is most appropriate for this operation?

Options:

- A. POST
- B. PUT
- C. DELETE
- D. GET

Correct Answer:

- D. GET

One-line Explanation:

GET is used to retrieve data from a server without modifying any existing resources.

Q.3 During application development, a developer encounters an error due to an incorrect or unavailable API endpoint, resulting in a "Not Found" response. Which HTTP status code most accurately represents this error?

Options:

- A. 404

- B. 500
- C. 403
- D. 400

Correct Answer:

A. 404

One-line Explanation:

HTTP 404 is returned when the requested endpoint or resource does not exist on the server.

Q.4 A developer writes a program that needs to create a new process and run a different executable file. Which pair of system calls fulfills this requirement in a Unix-like OS?

Answer: fork() creates a new process and exec() replaces it with a new executable.

Q.5 Which protocol is commonly used to secure data in transit over the internet by encrypting communication between systems?

Options:

- A. HTTP
- B. MTP
- C. SSL/TLS
- D. SMTP

Correct Answer:

C. TLS

One-line Explanation:

TLS (Transport Layer Security) encrypts data during transmission and is the modern, secure standard used for protecting data in transit.

Q.6 Assume that you have recently started working for an organization that is migrating to a cloud-based infrastructure. As part of IT governance, you are responsible for ensuring data security in the cloud environment. You are aware of various cloud deployment models and need to choose the most secure option for your organization's sensitive data. In the context of IT governance and cloud security, which cloud deployment model is typically the most secure choice for organizations with highly sensitive data?

Answer: Cloud Deployment Model - Security Private Cloud

One-line Explanation:

Private cloud provides the highest security for highly sensitive organizational data.

Q.7 After initializing a connection attempt to a remote server, consider that a client application sends a SYN packet to indicate the request for establishing a reliable connection. What is the expected response from the server, assuming it is available and prepared to accept the connection from the client?

Answer: The server responds with a SYN-ACK packet.

Q.8 A network administrator is tasked with selecting a secure wireless encryption protocol. Which of the following is considered the most secure for Wi-Fi networks?

Answer: WPA3 is the most secure wireless encryption protocol.

Q.9. Packet Routing / Firewall Bypass Attack Assume your computer's data was stolen by hackers through your network by mentioning the route to be used by the packets. This is done by the hackers to avoid security measures due to non-analyzing of source information instead of using a Packet filter firewall. What is this type of attack called?

Answer: This attack is called Source Routing Attack.

10. Computer Architecture - Accumulator to Memory

As a computer architect, you are designing the control unit of a CPU. You need to specify how data is moved from the accumulator to the memory. Which of the following best describes this operation?

Answer: Data transfer from accumulator to memory is done using a Store instruction.

11. Hybrid Cloud Connectivity

An enterprise is adopting a hybrid cloud model where they combine on-premise infrastructure with public cloud services to securely connect these environments and ensure reliability. What solutions should they implement?

Answer: Secure VPN or dedicated connectivity like Direct Connect ensures reliable hybrid cloud integration.

12. VPN for Remote Access

Assume that an employee at a multinational organization is working from home for some reason and needs to access the company's internal network to retrieve some files. Which type of VPN will be best suitable for this case?

Answer: Remote Access VPN is best suited for employees working from home.

13. DevOps - Continuous Deployment

The e-commerce company updates its website weekly by adding new features and fixes. Using an agile development approach, after each update, the new version is automatically deployed to the live environment. The practice involves automating the process of releasing software to production frequently and reliably. Which practice is being implemented?

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Options:

- a. Continuous Deployment - automates frequent and reliable releases to production
- b. Continuous Integration - focuses on merging code changes regularly to prevent integration issues
- c. Continuous Testing - ensures automated tests are run continually to validate code quality
- d. Monitoring - tracks application performance and user activity after deployment

Correct Answer:

- a. Continuous Deployment

One-line Explanation:

Continuous Deployment automates frequent releases to production, ensuring new updates are delivered reliably without manual intervention.

14. Routing Manipulation Attack

Hackers can manipulate data packets traveling across a network by inserting malicious routing information, allowing them to redirect traffic through their own systems. This method enables them to pull, extract, or edit data without detection. What type of attack is this called?

Answer: This type of attack is known as Man-in-the-Middle (MITM) attack.

15. OS Context Switching

During a context switch, consider that the operating system saves the state of the currently running process and loads the state of the next process to be executed. Which crucial CPU register responsible for loading the address of the next instruction to be fetched must be saved and restored during this operation?

Answer: The Program Counter (PC) register must be saved and restored.

16. I/O Bus and Device Controller

Suppose you want to design a bus system where input-output devices are connected to the computer via wires, and data transfer is handled electronically. The host controller sends messages to the device controller, which performs the operation. If the device controllers have a built-in cache for faster data transfer, which device will you choose?

Options:

- A. Enhanced Integrated Device Electronics A
- B. Enhanced Integrated Device Electronics B
- C. Enhanced Integrated Device Electronics C
- D. Enhanced Integrated Hard Driver

Correct Answer:

- B. Enhanced Integrated Device Electronics B (EIDE B)

One-line Explanation:

EIDE controllers include built-in caching to speed up data transfers between the host and storage devices.

17. Each party encrypts messages before sending them using a shared secret key, and messages are decrypted by the receiving party using the same key. If an attacker can send a specially crafted message to one party and analyze the

resulting encrypted output, which technique is most likely to compromise the secret key in this symmetric encryption system?

Options:

- A. Analyzing patterns in intercepted encrypted messages
- B. Comparing the crafted message's encrypted output with previously encrypted messages
- C. Performing a brute-force attack on intercepted messages
- D. Matching message lengths to guess the content

Correct Answer:

B. Comparing the crafted message's encrypted output with previously encrypted messages

One-line Explanation:

In a symmetric encryption system, a chosen-plaintext attack exploits the encryption of crafted messages to deduce the shared secret key.

18. Containerization Concept

A software development team is implementing containerization to streamline application deployment by encapsulating applications and their dependencies. Which characteristic defines containers in the context of software development and deployment?

Options:

- A. Virtual machines with a dedicated Hypervisor
- B. A logical separate network within the server
- C. Part of the same OS instance as the Hypervisor
- D. Virtual computers running under a Hypervisor

Correct Answer:

C. Part of the same OS instance as the Hypervisor

One-line Explanation:

Containers run isolated applications within the same OS instance, unlike virtual machines that require a dedicated hypervisor and separate OS.

19. WPA3 Configuration (Android)

Assume that a developer is tasked with configuring a high-security Wi-Fi environment for a sensitive government project using Android devices. The configuration includes:

Config SAE = Y

Config SUITEB = Y

Configure SUITEB192 = Y

In the given scenario, which configuration line is incorrect or redundant and might potentially weaken the desired security level?

Answer: Configure SUITEB192 = Y is redundant and may weaken compatibility without added benefit.

20. Cloud Elasticity / Resource Pooling

As part of a data center modernization initiative, the IT team is tasked with addressing storage challenges. The goal is to leverage existing storage devices from various vendors and integrate them with a next-generation storage solution. How does resource pooling play a crucial role in achieving elasticity in storage solutions for cloud computing?

Answer: Resource pooling enables dynamic allocation of shared storage capacity to meet demand.

21. OS Property - Application Compatibility

Consider you are a software developer working on a new application for a large corporation. Your application is expected to run on a variety of different computers with varying hardware specifications and operating systems. Which property of an operating system is most important for you to consider?

Answer: Resource Allocation ensures smooth application execution across varied systems.

22. Cloud Traffic Spike Handling

An e-commerce website experiences sudden traffic spikes during a flash sale. The platform needs to access resources to handle the increase without manual intervention. Which cloud features should be prioritized?

Answer: Auto-scaling and load balancing should be prioritized.

23. Shared Responsibility Model

Assume that a company migrates its application to a cloud-connected environment where the cloud provider handles infrastructure services while the customer develops and deploys the application. How does the shared responsibility model impact security management in this scenario?

Answer: Both the cloud provider and the customer share security responsibilities.

Q24. You are building a training module in MS PowerPoint and want to restrict navigation so that users must view each slide sequentially with narration, without freely skipping slides. Which setting provides the strictest control over slide navigation?

Options:

- A. Add hyperlinks between slides
- B. Use normal Slide Show mode
- C. Set up Slide Show as Browsed at a kiosk (full screen)
- D. Enable Presenter View

Correct Answer:

- C. Set up Slide Show as Browsed at a kiosk (full screen)

One-line Explanation:

"Kiosk mode" disables manual slide navigation and forces slides to advance only as designed, providing the strictest control.

Q.25 If a company wants to ensure secure and reliable email transmission for its cloud-based email service, which protocol should they implement?

Options:

- A. HTTP
- B. FTP
- C. SMTP
- D. POP3

Correct Answer:

- C. SMTP

Q26. To create a document template for repeated use, which file format should you save the document in?

Options:

- A..docx
- B. .pdf
- C..dotx
- D. .txt

Correct Answer:

C..dotx

One-line Explanation:

The .dotx format is specifically designed for Word document templates intended for repeated use.

Q27. An application relies on multiple hardware peripherals and requires strict isolation from other running containers to maintain data integrity and operational security. The development team uses a kernel-level configuration to allocate specific hardware resources to each container and restrict access to non-allocated devices. What is the most critical factor in maintaining operational security and preventing resource contention among containers?

Options:

- a. Ensuring that the host OS scheduler allocates CPU cycles equally to all containers
- b. Configuring the kernel to enforce strict resource isolation and limit peripheral access to only allocated devices.
- c. Allowing virtualization software to dynamically allocate additional memory as required
- d. Allowing containers to share network interfaces with individual access control lists

Correct Answer:

b. Configuring the kernel to enforce strict resource isolation and limit peripheral access to only allocated devices

One-line Explanation:

Strict kernel-level resource isolation prevents unauthorized hardware access and avoids resource contention between containers.

Q.28 Data is transferred between the CPU and a peripheral device through a shared bus controlled by device controllers to improve data transfer speed and reduce latency. The system uses controllers with dedicated high-speed buffers. Which of the following device controllers is most suitable for this purpose?

Options:

- A. Wireless Controller
- B. USB Controller
- C. SATA Controller
- D. Ethernet Controller

Correct Answer:

C. SATA Controller

Q29. Hackers attempt to intercept and manipulate data packets traveling across a network by inserting malicious code or routing information, allowing them to redirect traffic through their own system. This enables them to eavesdrop or alter data without detection. What type of attack is this called?

Options:

- A. Packet Sniffing
- B. IP Spoofing
- C. Routing Table Poisoning
- D. DNS Spoofing

Correct Answer:

C. Routing Table Poisoning

One-line Explanation:

Routing table poisoning redirects network traffic through the attacker's system, enabling interception or modification of data packets.

Q30. Question:

You are employed by a company that has recently suffered a security breach. As the IT manager, you are responsible for enhancing the security of the computer's operating system. Which property of an operating system should you prioritize to achieve this goal?

Options:

- A. Access Control
- B. User Interface
- C. Multitasking
- D. Virtual Memory

Correct Answer:

A. Access Control

One-line Explanation:

Access control manages permissions and restricts unauthorized access, making it critical for operating system security.

Q31. Jane wants to create a slide that illustrates a progression over time using icons, text, and blanks. She wants it to appear visually dynamic without manually animating each step. Which MS PowerPoint feature can help speed up this process?

Options:

- A. Use Slide Zoom feature to create dynamic progression
- B. Use a SmartArt process layout
- C. Apply the Morph transition across duplicated slides for a smooth effect
- D. Create a custom layout with background images and animation triggers

Correct Answer:

C. Apply the Morph transition across duplicated slides for a smooth effect

One-line Explanation:

The Morph transition automatically animates objects across slides, creating a smooth visual progression without manual animation.

Q32. During a context switch, the operating system saves the state of the currently running process and loads the state of the next process to be executed. Which CPU register, responsible for holding the address of the next instruction to be fetched, must be saved and restored during this operation?

Options:

- A. Stack Pointer
- B. General Purpose Register
- C. Program Counter
- D. Instruction Register

Correct Answer:

C. Program Counter

One-line Explanation:

The Program Counter stores the address of the next instruction, so saving and restoring it ensures the process resumes correctly after a context switch.

Q33. Two geographically separate offices, Office A and Office B, need to allow all users in Office A to securely access resources in Office B and vice versa, as if they were on the same local network. Which VPN topology is most appropriate for this scenario?

Options:

- A. Remote Access VPN
- B. Site-to-Site VPN
- C. Client-to-Server VPN
- D. Peer-to-Peer VPN

Correct Answer:

B. Site-to-Site VPN

One-line Explanation:

A Site-to-Site VPN connects entire networks across locations, allowing users in both offices to securely access each other's resources as if on the same LAN.

Q34. A development team is designing a new RESTful API to manage a product catalog for their e-commerce platform. They need to create an endpoint that retrieves details of a specific product using its unique product ID. Which HTTP method and URL structure best align with RESTful principles for this operation?

Options:

- A. POST/products/{productid}
- B. GET/products/{productid}
- C. PUT/products/{productId}
- D. DELETE/products/{productid}

Correct Answer:

B. GET/products/{productid}

One-line Explanation:

The GET method is used in RESTful APIs to retrieve data, and including the product ID in the URL specifies the resource to fetch.

Q35. A containerized application is consuming massive memory, causing performance issues. What is the recommended approach to address this problem effectively?

Options:

- A. Increase the memory limit for the container
- B. Use a larger base image for the container
- C. Reduce the number of replicas of the container
- D. Analyze the application code for memory leaks

Correct Answer:

D. Analyze the application code for memory leaks

One-line Explanation:

Memory issues are best resolved by identifying and fixing leaks in the application code rather than just increasing container resources.

Q36. A cloud infrastructure requires running multiple isolated environments on a single physical server. Which technology is primarily used to achieve this?

Options:

- A. Containers
- B. Virtual Machines (VMs)
- C. Serverless computing

Correct Answer:

B. Virtual Machines (VMs)

One-line Explanation:

Virtual machines provide full isolation for multiple environments on a single physical server, making them ideal for securely running separate workloads.

Q.37 An internet connection is established by a user, connecting devices like a computer, smartphone, and smart TV to a router. Which OSI model layer is responsible for transmitting Ethernet, cable, and Wi-Fi signals between these devices?

Options:

- A. Physical Layer
- B. Network Layer
- C. Data Link Layer
- D. Application Layer

Correct Answer:

- A. Physical Layer

One-line Explanation:

The Physical Layer handles the transmission of raw electrical, optical, or wireless signals over media like Ethernet cables and Wi-Fi.

Q39. In a PivotTable, which feature allows you to create custom calculations and summaries based on the existing data?

Options:

- A. Calculated Fields
- B. Calculated Items
- C. Slicers
- D. Grouping

Correct Answer:

- A. Calculated Fields

One-line Explanation:

Calculated Fields let you define custom formulas in a PivotTable to perform calculations on the summarized data.

Q40. Data packets are transmitted between routers and switches to ensure proper delivery across different network segments. Which OSI model layer is responsible for establishing logical paths and routing data packets between different network segments?

Options:

- A. Data Link Layer
- B. Network Layer
- C. Transport Layer

D. Session Layer

Correct Answer:

B. Network Layer

One-line Explanation:

The Network Layer determines logical addressing and routing to deliver packets across multiple network segments.

Q41. In a cloud computing environment, which security measure helps protect user accounts by requiring two different types of identification before granting access?

Options:

A. Firewalls

B. Encryption

C. Multi-Factor Authentication (MFA)

D. Antivirus Software

Correct Answer:

Correct Answer:

C. Multi-Factor Authentication (MFA)

One-line Explanation:

MFA enhances account security by requiring multiple forms of verification, such as a password and a one-time code.

Q42. A small business uses ring network topology. If a single workstation goes down, what impact will it have on the network?

Options:

A. Minimal impact; network reroutes traffic

B. Network becomes isolated

C. Network slows down

D. Network breaks completely

Correct Answer:

D. Network breaks completely

One-line Explanation:

In a ring topology, each device is connected in a closed loop, so the failure of one workstation disrupts the entire network.

Q43. What is the process of converting data into a secure format to prevent unauthorized access?

Options:

- A. Encryption
- B. Authentication
- C. Authorization
- D. Compression

Correct Answer:

- A. Encryption

One-line Explanation:

Encryption transforms data into a coded format to ensure that only authorized parties can access it.